4. **Collaborating Institutions/Agencies/Industries (if any):**
   i. National Institute of Disaster Management (NIDM) (Government Agency)
   ii. Tata Consultancy Services Limited (TCS) (Industrial Organisation)
   iii. Indian Institute of Technology Indore (Academic Institute)
   iv. Forest Research Institute, Dehradun
   v. Indian Institute of Remote Sensing, Dehradun
   vi. VARISHT- (Voluntary Association of Researchers in Indigenous Knowledge, Spirituality & Science, Technology and Homeland Security).
   vii. ESRI India
   viii. Sikkim State Disaster Management Authority
   ix. Kerala State Disaster Management Authority
   x. CWRDM, Govt. of Kerala
   xi. Sarathy Geotech Engg Services Ltd.

5. **Duration (max Upto 3 years): -** 03 Years

6. **Total Budget: 3,43,29,660**

   DST Share: Rs. 2,99,99,660

   External Agency Share: 43,30,000

7. **Objectives (***In bullet form***):**
   The goal of this project is to develop an intelligent responsive system based on a comprehensive knowledge database built on disasters, traditional knowledge of mountain communities, real time monitoring sensors, spatial information systems, and auxiliary sources of data focusing on multi-hazard early warning and management in hilly regions of India. More specifically the objective is to support AI based tools for knowledge extraction and decision making for efficient coordination in disaster management and response. Thus, the following are the broad objectives of the project

   - Implement frameworks for obtaining static and dynamic data from different source types like hazard monitoring sensors, laboratory model test results (rainfall threshold and critical acceleration), traditional knowledge from local communities, spatial information sources, and web sources providing weather, news, and social media reports.
   - Design and development of Integrated Wireless Sensor Probe (IWSP) for capturing Multihazard parameters.
   - Develop automated techniques for processing, refining and fusing multimodal data obtained from these sources to create a hazard data lake.
   - Develop AI and ML based models for hazard mapping and susceptibility indexing, knowledge extraction for event prediction, early warning, real time situation

awareness, event identification, and timeline generation.

- Develop a unified evaluation framework for AI based models targeted towards the above-mentioned goals.
- Replicate real field scenarios in the laboratory based on the available geological/geotechnical data and topography of a vulnerable site and simulating rainfall and earthquake conditions to validate early warning of landslides due to combined effect of rainfall and earthquakes
- Develop appropriate use cases for leveraging project outcomes, identification of evaluation markers, establishment of technology at laboratory scale and deployment of proof of concept. Focus area will be selected sites of Uttarakhand.

8. Novelty/Innovative Elements/S&T Components of the Project:

The major novelty of the project is to develop a comprehensive data collection framework for capturing relevant multimodal hazard data from diverse sources like sensors, local communities, GIS, and web sources to create a data lake that is subsequently leveraged by the administration for coordinating various disaster management tasks, as well as providing user-centric services. Furthermore, it also aims to provide benchmarking and evaluation support for researchers and developers to improve its quality and scope. The major innovative components include:

1. Creating a hazard data lake with heterogeneous data collected from diverse sources to facilitate analytics and AI based support tools.
2. Developing an Intelligent Wireless Sensor Probe (IWSP) which is Self-sustaining, Context-aware, and offers Adaptive Service Delivery
3. Providing holistic solutions for coordinating various disaster management tasks leveraging domain specific expertise from various disciplines, which otherwise works in silos.
4. Exploiting traditional knowledge gained from local communities for better planning and decision-making based on location and community-specific requirements and cultural practices.
5. ML and AI based decision models for detection and early warning of multi-hazards
6. Physical model tests to validate a real field scenario considering geological/geotechnical parameters and rainfall & earthquake vibrations
7. Developing a novel automated benchmarking and evaluation platform for AI based developers and researchers working on various disaster management tasks.

9. Details of the prototype developed/Initial work done in the proposed Development of Technology

<u>IIT Patna</u>

IIT Patna would be working on 4 major components: raw data collection from static and dynamic data sources, benchmark dataset creation, development of AI based models for information extraction, retrieval and filtering from multimodal data, and a framework for unified model evaluation.

The PI's research group has worked significantly on the first 3 components. The group has developed a big data framework for collection and storage of large scale streaming multimodal data collected from social media feeds like Twitter, Facebook and Reddit. The group is also working on collection of image and video feeds from web sources as well as capture devices.

Further the group has also worked significantly on AI based models for information retrieval and filtering of relevant multimodal social media feeds related to specific tasks, like infrastructure damages in case of disaster. The IIT Patna group will be actively collaborating with Tata Consultancy Services (TCS). TCS has done relevant research work in developing machine learning and NLP based models for knowledge extraction, event identification, timeline generation and similarity analysis etc. These models have been used to analyse and suggest preventive measures for the industrial incidents and disasters. These advances by both groups would be rigorously used for further development in the project.

UPES Dehradun

Forest fire is always a major issue in many parts of India during the summer season every year. In the month of Feb-March 2021, due to the record high temperatures, the Similipal forest fire was a major disaster affecting the rich flora and fauna of this protected region. The UPES group used multi-sensor space-borne satellite data from Sentinel-1, Sentinel-2, and Landsat-8 for forest fire assessment of study area Similipal National Park, Odisha, India. It is found significant results in the identification and evaluation of the burnt area and severity label of forest fire. In this work the ground truth data is provided by Dr. Dheeraj Gupta , Scientist, TFRI, Jabalpur and FRI Dehradun.

Further the UPES group also working on landslide prediction and developed hybrid machine learning model for Uttarkashi, Uttarakhand. In this work, the Rough Set theory coupled with five different models namely Bayesian Network (HBNRS), Backpropagation Neural Network (HBPNNRS), Bagging (HBRS), XGBoost (HXGBRS), and Random Forest (HRFRS) were taken into account. The database for the models development was prepared using fifteen conditioning factors that had 373 landslide and 181 non-landslide locations. The data used in this research is taken from the open-access platform of geological survey of India (GSI). The published work link is https://www.nature.com/articles/s41598-022-22814-9

Another work are published on semantic segmentation of semantic segmentation of PolSAR data using advanced deep learning model for LULC purposes. The link of the paper is https://www.nature.com/articles/s41598-021-94422-y

IITR

The PI from IIT Roorkee work has focused on the study of vernacular settlements in disaster context. He is currently working on an international collaborative project jointly funded by ICSSR(India)-JSPS(Japan). The project titled" An Integrated framework for integrating Climate Change Adaptation and Disaster Risk Reduction" is focussed on

understanding the responsive nature of Vernacular settlements in Himalayan region. They have recently completed Yonmengeiki exercises with the local communities. The upcoming project that has been sanctioned for funding under DST-SHRI call is focussed on developing “A comprehensive GIS-based toolkit for investigating the cultural routes of Bhotia's trade routes in Uttarakhand”. Very Recently, PI, has been sanctioned a funded research project focussed on mountain communities under DST-SHRI scheme titled- “A comprehensive GIS-based toolkit for investigating the cultural routes of Bhotia's trade routes in Uttarakhand.

In this project he will be working in collaboration with an NGO, VARISHT- (Voluntary Association of Researchers in Indigenous Knowledge, Spirituality & Science, Technology and Homeland Security). Previously, as a team, they have won second prize in AWS Hackethon for the project titled Predicting Thunderstorms using Machine learning. Historical Data was collected over the years from open sources for wind-speed, wind direction, temperature, precipitation, and humidity from Northern and Southern India and parts of the USA. Part of historical data was used to create the predictive model while the remainder of the data was used to validate the model (70/30). By recording and correlating the atmospheric conditions at the time of lightning strikes during thunderstorms, our Machine Learning Model predicts whether a dangerous lightning strike will occur within the next 3 hours. The link for this project is as follows. Thunderstorm Prediction using ML | Devpost. The team also presented the similar broader idea in various forums including a recent paper in the technical session at the 5th World Congress of Disaster Management, IIT Delhi, 24-27 November 2021.

Amrita Vishwa Vidyapeetham

Amrita has been engaged in landslide related research since 2006 and has more than a decade long experience in landslide monitoring and early warning systems. Landslide research work was initiated to study rainfall induced landslides of Munnar, Western Ghats in India. As part of this research work more than 150 sensors were installed on an unstable hill to monitor the possibility of landslides. This system was integrated using a WSN architecture, but was later upgraded to an IoT architecture. The people of Anthoniyar colony living at the base of this unstable hill were the primary beneficiaries of this system. This system has generated several successful early warnings there by evacuating the people during hazardous situations as seen during monsoon seasons.

Early warning’s were successfully issued in July-2009, August-2011, August-2013, July,August-2018, August-2019, August-2020, July-2021, July-2022 to the Munnar community through the Government. After the success of this system, a similar IoT system was deployed in the thickly populated Gangtok town in Sikkim. This hill was instrumented with more than 250 sensors, the electronics and other subsystems were also optimized to perform in himalayan conditions.

Amrita is also involved in detailed and inclusive community scale disaster resilience programs. To enhance the disaster resilience Amrita has also developed crowdsourced

applications such as (a) Landslide Tracker, a collaborative mobile application developed as part of LANDSLIP project to track the landslides using crowdsourcing approaches, (b) Amrita Kripa mobile application for real-time disaster management, and c) Social Media Mining Platform for Multi Hazard Event Detection, with specific focus on rescue, and relief operations in an inclusive manner with diverse stakeholders. Capacity building programs have been initiated to enhance the community engagement and empowerment for effective disaster risk reduction and for enhancing the community level disaster resilience. As part of this research two patents are also awarded.

- US Patent: Method of reducing DoS attacks using voice response in IoT systems; Patent number: US 11,343,273 B2; Inventors: Sangeeth Kumar, Venkat P. Rangan, Maneesha Vinodini Ramesh
- Indian Patent: Adaptive Energy Management System and Method for Real Time Landslide Detection",; Patent number: 369608; Inventors: Dr. Rekha Prabha and Dr. Maneesha Vinodini Ramesh

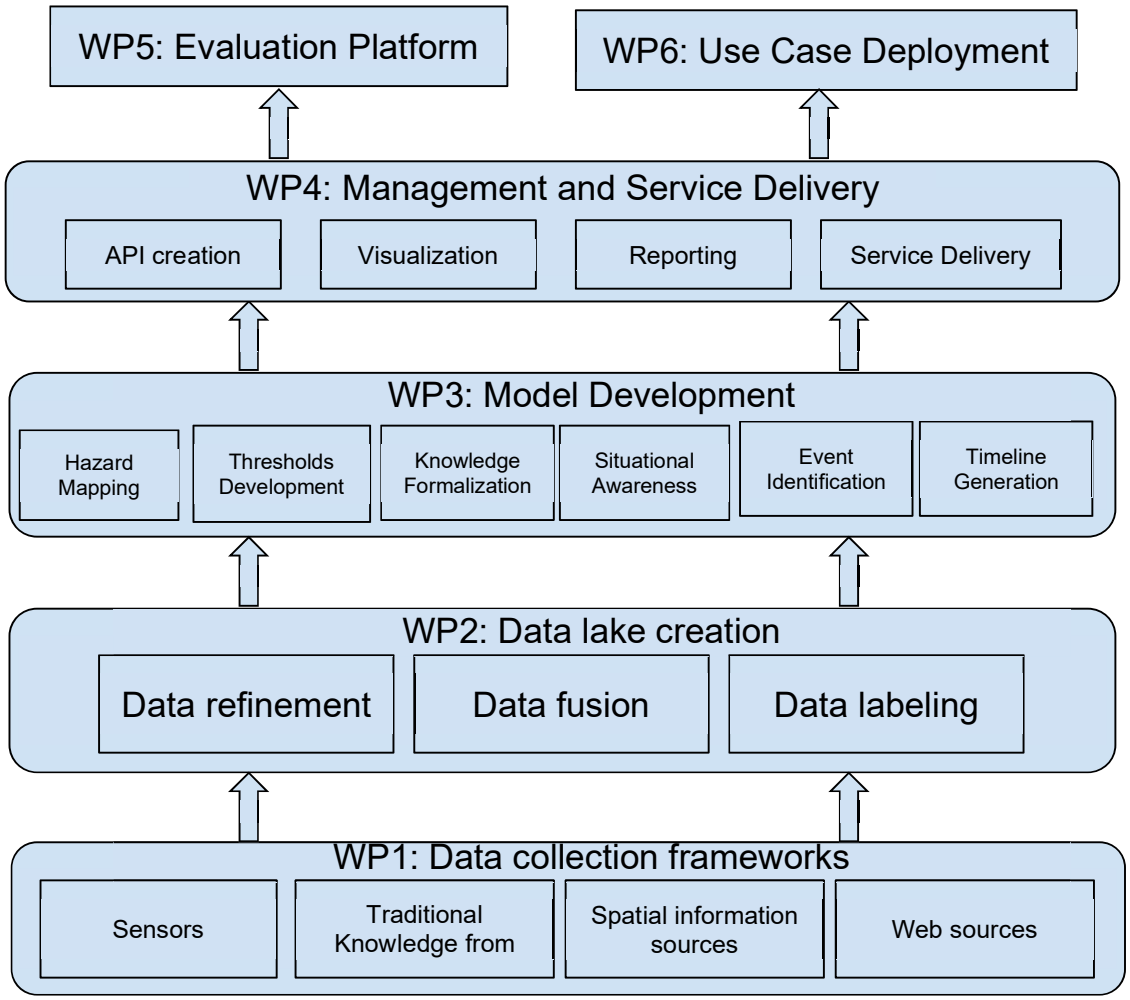
<u>NITK</u>

- NITK has developed a MATLAB based platform to evaluate the seismic hazard using state of the art deterministic and probabilistic approaches and option to generate landslide hazard map for a region by integrating 12 geological parameters along with rainfall intensity and earthquake acceleration expected in the region.
- NITK has already developed, tested and demonstrated a landslide real time monitoring system using landslide flume recreating the field scenario including rainfall simulation.

10. TRL (at present) (1-9 scale): 4

11. Methodology:

The entire project is divided into 6 work packages, each with a specific set of objectives and deliverables. The details of the work package is represented using a diagram as shown below. We elaborate each of the work packages in the project detail section.



Methodology flow chart

12. Outcome/ Deliverables and their Expected Impact (*In bullet form*):

Following are the major outcomes of the project:

A. A new integrated product to be used in large scale multi hazard scenarios for automated data collection from different source types like hazard monitoring sensors, laboratory model test results (rainfall threshold and critical acceleration), traditional knowledge from local communities, spatial information sources, and web sources providing weather, news, and social media reports for multihazard parameter capturing, hazard mapping, real time situational awareness, event identification, and timeline generation of various events in a past disaster

B. Technical reports on novel processes proposed for hazard mapping, real time data collection, multimodal data fusion, creating benchmark data lake from heterogeneous sources, formalism of knowledge gained from local communities, situational

awareness, and event identification and timeline generation.

C. Development of a WebGIS portal and deployment of AI models through the portal
D. Proof of concept deployment for use case scenarios with the technical report on the proof of concept deployment, and evaluation and field report.
E. A unified evaluation system for testing AI based models related to task specific information extraction, retrieval and filtering, event detection, and timeline generation. This can be used to keep track of the best performing model for each relevant task used in the proposed system.
F. Proof of concept for real-time monitoring and early warning of landslides due to the combined effect of varying rainfall intensity and earthquake vibrations
G. A data lake with set of labeled multimodal benchmark datasets for training models related to different disaster management tasks
H. Trained manpower with knowledge in IoT, AI and Machine Learning, and Big data technologies

13. Target Beneficiaries & Benefits to the country:

Following are the target beneficiaries

- Government administration for overviewing and coordinating disaster management tasks, real time situational awareness, damage prediction and decision making
- Voluntary organizations like VARISHT for need and requirement prediction for post disaster relief and support
- Real-time monitoring and multi-level early warning against landslides can be used by the National Disaster Management Authority, Indian Railways, National Highway Authority of India, State Govt. Disaster Management Authorities, and other organizations like Tata Coffee Ltd., MRPL etc
- Researchers, who can use the benchmark data as well as the evaluation platform for training and evaluation of certain task specific models
- Local Communities living in Mountain regions, and their respective stakeholders

14. Brief Technical Details (*300 words*), giving Justifications for the Project, the underlying Scientific Basis and the Methodology:

The disasters that hit India resulted in an average annual loss (AAL) of several hundred billion dollars, as estimated by the United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), which the World Metrological Organization (WMO) referred to in its report. In 2020, China recorded the highest loss, at $238 billion, followed by $87 billion in India and $83 billion in Japan. The highest AALs were associated with drought, the WMO said.

Every year, due to disasters, thousands and thousands of people face death and hardships in India and around the world, which is a matter of great concern, especially given their impacts on lives and livelihood. Certainly, there is scope to put certain mechanisms in place in order to create a robust disaster management system supported by a strong data lake to take the timely and appropriate response. Technology has the potential to transform disaster management. The success of this project could encourage both state-wise and national

agencies to adopt the use of Drones, Artificial Intelligence, Big Data, Internet of Things (IoT) coupled with the traditional knowledge of cultures and practices of local communities for making our response and recovery (relief) systems faster and more effective. For states such as Uttarakhand that rely heavily on tourism, a faster recovery from disasters could rejuvenate its economy much faster to reduce losses.

15. Cost-Benefit (C-B) Analysis)/ Cost-Economic impact Analysis Report (Details of how this technology will impact the society in economic terms)

Disaster management requires proper infrastructure for early warning and detection along with dedicated manpower and other resources for both decision making and on the ground activities for preparedness, response, as well as post disaster recovery in natural hazards. While decision making is based on both past experience as well as current ground situation reports, obtaining both long term experience as well as the current situation during any disaster may be challenging. According to the figures published by Statista research department in 2022, India has spent more than 300 billion Indian Rupees for relief in consecutive years 2016 and 2017. As disaster situations demand quick response and rehabilitation efforts by the administrator, quick and proper decisions in terms of resource distribution and workforce mobility is required. Improper decisions based on insufficient or incorrect information may incur huge wastage and loss to the exchequer. According to an annual report issued in 2012 in US by their Justice Department's Disaster Fraud Task Force it was mentioned that in cases related to Hurricanes Katrina, Rita and Wilma alone, the task force through FY 2011 prosecuted 1,439 individuals in 47 federal districts throughout the country for improper usage of disaster relief funds.

The situation is no different in India where several media reports were published regarding inaccuracies in fund distribution after several natural disasters. The major goal of the data-driven techniques for disaster management is to facilitate the administration in decision making through automated means by gaining quick knowledge from the past disasters as well as obtaining a more precise view of the current situation post disaster. This would help in better decision making and would mitigate the risks involved in distribution of relief funds, in terms of wastage, scams and fraud.

16. Role of Industry in the proposed project:

Industries that work on disaster related systems may play a very important role in the project by improving its techno-commercial aspects. The role of the industry would be to ensure that the developed product is more market driven and meets the industry standard in terms of the quality. This would enhance the commercialization prospects and reduce the time to market. Further, partnering with the industry to leverage their expertise in commercial system development is expected to mitigate several unforeseen risks in the project. Hence the team has partnered with Tata Consultancy Services (TCS). TCS has been actively involved in industrial disaster management related research and would be playing an important role in

system requirement specification as well as in the design and development phase. TCS has done relevant research work in developing machine learning and NLP based models for information and knowledge extraction, identifying events, entities as well as frequent patterns from historical incident reports, representing and formalizing the extracted knowledge to facilitate further reasoning, generating timelines and identifying similarity incidents etc. These models have been used to analyze and suggest preventive measures for the industrial incidents and disasters. Furthermore, more importantly, TCS is providing support in terms of manpower (through paid internships) to be leveraged for the project.

17. Technical specifications of the final deliverables: -

Following are the final deliverables and their specifications:

1. Data collection framework
    1. Real-time data collection from the sensors deployed in the field.
    2. Supported static data sources should include historical incident and damage reports, images and videos from news and weather repositories, text from multilingual news articles, and past available social media feeds.
    3. Supported dynamic data sources should include downloadable images from various news and meteorological sites, texts from news articles, published incident reports, and social media feeds like Twitter, Reddit and Facebook.
    4. Supported languages for text analysis should be English
    5. Should support standard image and video formats, html and xml formats, and json format for social media feeds
2. Analytics framework
    1. Real-time warnings
        i. Regional and site specific threshold
        ii. Multiple levels of warning
    2. Situational awareness in terms of
        i. Affected location names
        ii. Damage score
        iii. User requests and needs
        iv. Injuries and deaths
        v. Response to evacuation
    3. Event identification and Timeline generation
        i. All major natural and manmade disasters in the last 5 years
        ii. Graphical visualization
3. Technical framework
    1. Parameters and protocols that would be used to develop a comprehensive response and recovery system for Disaster Management.
    2. A laboratory scale AI based comprehensive response and recovery system for disaster management that would be finalized after it has been tested through simulation studies.
    3. Integrated technology/hardware system called as Artificial Intelligence (AI) based

Comprehensive Response and Recovery System for Disaster Management. This integrated technology system would involve seamless integration of High Resolution Cameras, Unmanned Aerial Vehicle (UAV), High Speed Data Processor, Relays and Communication Systems for multidirectional flow of data for faster and effective responses to disasters by various concerned agencies.

4. WebGIS portal and deployment of AI models through the WebGIS portal for user access.

4. Data repositories
   1. Separate repositories for sensor, spatial, image and video data, web articles and social media feeds
   2. Suitable anonymization of all data
5. Heat maps and visualizations
   1. Sensor data from the deployment site
   2. Risk mapping based on cultural perceptions and community based responses
   3. Hazard, exposure and vulnerability maps
   4. Resource mapping
6. Unified evaluation platform
   1. Provides data objects for labeled data compatible with python libraries
   2. Should support codes written in python for task specific information filtering, event identification and timeline generation
7. Detailed assessment of earthquake hazard using deterministic and probabilistic approaches
8. Geological and geotechnical considerations for preparing Landslide vulnerability maps induced from both rainfall and earthquakes considering Frequency ratio method and Newmark analysis
9. Lab-based physical model for early warning of landslides with combined effect of rainfall and earthquakes

18. Host institute contribution for the execution of the project:

Following are the institute wise contribution to execute the project:

Activities and responsibilities of each institute

Work Package	Activity & Task	NITK	IITR	Amrita	IITP & TCS	UPES
WP1: Development of data collection and Ingestion Framework Lead:	IoT Edge Node Development			X		
	Sensor Data			X		X
	Traditional knowledge from local communities		X			
	Spatial Data	X		X		X

	Multimodal data from web sources				X	
WP2: Data refinement, pre-processing and labelling Lead:	Sensor data			X		X
	Edge Computing			X		
	Traditional knowledge data		X			
	Spatial data	X		X		X
	Web source data				X	
WP2: Data alignment and fusion Lead:	Automated techniques for multimodal data alignment and fusion			X	X	
WP3: Model Development Lead:	Hazard mapping	X		X		
	Preparedness indexing	X		X		X
	Knowledge extraction from community data		X		X	
	Situational awareness and damage assessment		X	X	X	X
	Major event identification from historical data			X	X	
	Disaster specific activity timeline generation			X	X	
	Real-Time Event Detection Models	X		X		X
	Real-Time Early Warning Models	X		X		X
WP4: Disaster Management & Services Lead:	WebGIS & Visualisation		X		X	
	Warning Dissemination					X
	API creation				X	X
	Report Generation				X	

WP5: Benchmarking and evaluation Lead:	Identifying critical features for assessing community specific vulnerabilities and risks		X			
	Identifying critical features for assessing post disaster response and recovery		X			X
	Identifying parameters for hazard prediction	X		X		
	Platform for automated evaluation of AI based models on damage and need assessment, event detection and timeline generation			X	X	
WP6: Use case development and proof of concept deployment Lead:	Proof of concept	X	X	X	X	X
	Use cases on hazard mapping and preparedness indexing	X		X		
	Use cases on community based studies		X			
	Use cases on damage and need assessment, and task planning				X	
	Use cases on assessing post disaster response and coordination					X
	IPR and Commercialization plans	X	X	X	X	X

19. Any other relevant information:

PI of NITK is consultant to Indian Naval Academy, Southwestern Railways and Tata Coffee Estate who have expressed their interest in early warning devices as end user.
PI of UPES is associated with NISAR (NASA-ISRO Synthetic Aperture Radar) mission and completed a project under ASAR sponsored by SAC, ISRO Ahmedabad.

2. Detailed Project Proposal